Name: Hemaksh Breja

SRN: PES2UG24CS191

Section: 5D

Lab 3: Architectural Pattern Selection & Justification

Architecture:

We chose the Layered Architecture (Presentation, Business, and Data layers) for the Academic Elective Bidding and Allocation System.

Reason 1: it exactly matches the System's Natural Processing Stages

The system's workflow naturally separates into distinct stages: student/registrar interaction (Presentation), bid processing and allocation logic (Business), and persistent storage of bids, capacities, and results (Data). A layered structure keeps these concerns cleanly separated, making the Bid Submission, Prerequisite Validation, and Allocation Engine components easy to design, implement, and test independently.

Reason 2: it is easier maintenance and extensibility

Business rules such as the tie-breaker policy (CGPA, then submission timestamp) or prerequisite checks are likely to change across academic terms. With a layered architecture, these rules live in the Business layer and can be updated without touching the Presentation layer (student/registrar portals) or the Data layer (database schema), reducing regression risk.

More Secure:

The layered structure enforces that the Presentation layer never accesses the Database Component directly all requests are routed through the Business layer (Bid Submission, Allocation Engine). This supports NFR-002, ensuring a Student can only retrieve their own bid records while the Academic Registrar retains full visibility, since access-control checks are centralized in the Business layer rather than scattered across UI code.

Benefit:

Isolating the Allocation Engine Component within its own layer allows it to be optimized and scaled independently (e.g., batch-processing bids in memory before writing final results) without being constrained by UI rendering or database I/O patterns. This directly supports NFR-001, which requires the allocation run for up to 5,000 students to complete in under 2 minutes.